

1. EXECUTIVE OVERVIEW

This report summarizes the successful engineering, simulation, and validation of a high-fidelity Digital Twin representing a 3kV DC Traction Substation (TSS). The core objective of this modeling project was the integration of a 3 MW Battery Energy Storage System (BESS) designed specifically for localized Peak Shaving and dynamic Voltage Support. Engineered in the PSCAD environment, this digital twin accurately replicates complex electromechanical transients, severe grid fault dynamics, and nonlinear traction load behaviors. The validated model serves as a foundational proof-of-concept for upgrading our legacy traction networks to support modern, high-demand rolling stock while extending the operational lifespan of existing substation infrastructure.

2. SYSTEM ARCHITECTURE & METHODOLOGY

The digital twin was constructed using a bottom-up, physics-based methodology to ensure strict adherence to real-world electrical behavior:

- **Main Grid Supply:** Modeled as a robust 132 kV AC transmission grid feeding a heavy-duty 12-Pulse Rectifier. This topology establishes a nominal 3.0 kV DC Link while realistically simulating the characteristic 600 Hz harmonic ripple.
- **BESS Infrastructure:** Integrated a 1.2 kV Lithium-Ion Battery pack interfaced to the 3.0 kV DC link via a Bidirectional Interleaved DC-DC Converter. This topology was selected to minimize current ripple and thermal stress on the battery cells.
- **Dynamic Traction Load:** Standard, idealized resistor models were discarded in favor of a 3 MW Constant Power Load (CPL). This accurately simulates the critical "Negative Incremental Impedance" effect characteristic of modern traction inverters, where a drop in line voltage causes a proportional increase in current demand.
- **Train Input Filter:** The traction load was paired with a physically accurate LC input filter (5 mH Line Choke and 10 mF DC-Link Capacitor), complete with a localized Pre-Charge Resistor circuit and a Main High-Speed Circuit Breaker (HSCB) to accurately model inrush current phenomena.

3. CONTROL SYSTEM DESIGN

To manage the complex interplay between the BESS, the main grid, and the train, a highly autonomous control architecture was deployed:

- **Master State Machine:** Developed a logic controller to autonomously switch the BESS between Charge Mode (capturing regenerative braking energy via Buck

operation) and Discharge Mode (providing active voltage support via Boost operation during heavy acceleration).

- **Cascaded Control System:** Implemented a robust dual-loop architecture. The outer Voltage PI Loop aggressively regulates the substation DC Link to maintain 3.0 kV. An inner Current Loop Limiter ensures battery discharge never exceeds a safe thermal ceiling of 2.0 kA.
- **Traction Soft-Start Ramp:** To prevent the 3 MW CPL from causing catastrophic instantaneous voltage collapse upon startup, a First Order Lag (Real Pole) power ramp was programmed into the train's inverter logic. This forces a gradual power draw, granting the grid and BESS time to dynamically respond and stabilize.

4. HARDWARE PROTECTION & FAULT RECOVERY

The digital twin incorporates a comprehensive, autonomous safety architecture designed to withstand and recover from dead-short DC link faults:

- **"Two-Headed Dragon" Mitigation:** Specifically addressed the unique fault contribution of the BESS, where the DC-DC converter's anti-parallel diodes feed a fault even when the IGBTs are actively blocked.
- **ANSI-Standard Protection:** Engineered a custom mechanical protection scheme utilizing an SR Flip-Flop (acting as a Latching Relay). This brain monitors the system and triggers purely DC HSCBs—configured physically to "open at any current" to prevent zero-crossing wait errors—the moment fault currents exceed 5 kA.
- **Autonomous Auto-Recloser:** Implemented an automated recovery sequence using timed step-logic. Once the fault is cleared, the system resets the latching relay memory and smoothly restores grid operations without human intervention.

5. SIMULATION CHALLENGES & RESOLUTIONS

Several critical mathematical and software-specific anomalies were identified and resolved to achieve simulation stability:

- **Matrix Stability (Divide-by-Zero):** Solved simulation-crashing infinite current demands during CPL start-up (when $V=0$) by implementing a hard clamp limit, restricting the denominator voltage calculation to a safe minimum of 2.0 kV.
- **Data Type Mismatches:** Mitigated inherent PSCAD logic errors (Logical vs. Real data types) in breaker control matrices by abandoning standard switches in favor of pure arithmetic logic routing (e.g., 1.0 - Signal for active-low operations).

SUBJECT: Executive Summary: PSCAD Digital Twin of 3kV DC Traction Substation with Integrated 3MW BESS ----- PSCAD 4.6.2 -----

- **Integral Windup:** Documented a severe overvoltage transient upon auto-reclosing. This was diagnosed as Integral Windup within the PI controllers, where the error accumulated during the breaker open-state (0 V) resulted in an aggressive overcompensation spike. This finding highlights the strict necessity for implementing PI Anti-Windup logic in the physical BESS PLC firmware before deployment.

6. REFERENCES & STANDARDS

The digital twin and component sizing were strictly benchmarked against current international railway standards:

- **EN 50163 Compliance:** The entire control methodology was engineered to guarantee compliance with the European standard for railway supply voltages, ensuring the steady-state catenary voltage never permanently drops below the U_{min1} threshold of 2.0 kV under maximum load.
- **Empirical Component Sizing:** The 10 mF train capacitor was sized per the industry standard 3-4 mF/MW ride-through rule of thumb. The 5 mH line choke was mathematically derived to limit the fault current rate of rise (di/dt) to a maximum of 700 A/ms, remaining safely within the mechanical extinguishing capabilities of modern DC HSCBs.

Conclusion:

The completed PSCAD model validates the operational resilience of a BESS-supported 3kV DC traction network. The digital twin provides a reliable sandbox for future hardware-in-the-loop (HIL) testing and final control firmware development.